ICC-Tutorial: Worksheet for Safety Verification of Neural Networks using Starsets

Lipsy Gupta[1], Ratan Lal[2], Pavithra Prabhakar[3]
[1]Kansas State University
[2] Northwest Missouri State University
[3] University of New Mexico

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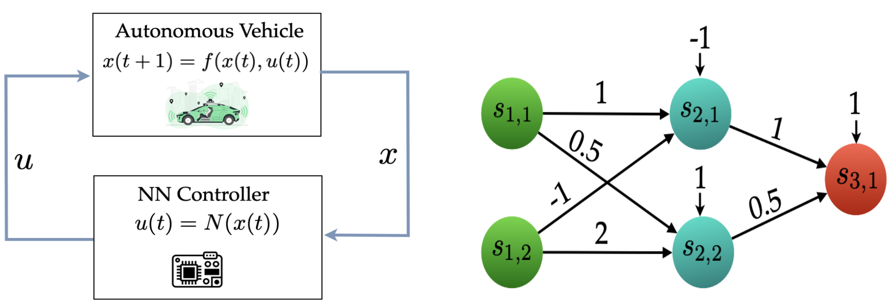
Worksheet 1: Computing outputs of the cloed loop- NN and the dynamics

Goal

- Evaluate a neural network on a given state.
- Use the NN output as a control input.
- Apply one step of linear system dynamics.
- Repeat the process and write a short execution trace.



(a) NN-controlled dynamical system (b) Neural Network $\mathcal{N}_1$

Figure 1: Closed-loop system and neural network controller

Plant dynamics. The system evolves as

$$x(t+1) = Ax(t) + Bu(t),$$

where

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Controller. The control input is given by a neural network:

$$u(t) = \mathcal{N}(x(t)).$$

Use the network $\mathcal{N}$ shown in Fig. 1b (2 inputs, 2 hidden, 1 output). The network is $\mathcal{N} = \{(W_1, b_1), (W_2, b_2)\}$ with

$$W_1 = \begin{bmatrix} 1 & -1 \\ 0.5 & 2 \end{bmatrix}, \quad b_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \qquad W_2 = \begin{bmatrix} 1 & -0.5 \end{bmatrix}, \quad b_2 = 1.$$

Initial state at $t = 0$.

$$x(0) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Task 1: Compute the NN output $u(0)$

1. Compute the first-layer pre-activation:

$$z_1^{(0)} = W_1 x(0) + b_1 = \begin{bmatrix} 1 & -1 \\ 0.5 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 6 \end{bmatrix}.$$

2. Apply ReLU coordinate-wise:

$$y_1^{(0)} = \mathrm{ReLU}(z_1^{(0)}) = \begin{bmatrix} \mathrm{ReLU}(-1) \\ \mathrm{ReLU}(6) \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}.$$

3. Compute the output-layer value:

$$z_2^{(0)} = W_2 y_1^{(0)} + b_2 = \begin{bmatrix} 1 & 0.5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} + 1 = 3 + 1 = 4.$$

4. Apply ReLU and write down the control input:

$$u(0) = \mathrm{ReLU}(z_2^{(0)}) = \mathrm{ReLU}(4) = 4.$$

Answer: $\boxed{u(0) = 4}$

This flow of values in the NN is shown in Figure 2

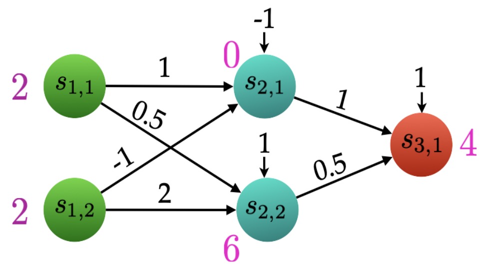


Figure 2: Neural network $\mathcal{N}$ with values illustrated

Task 2: Plant update to $x(1)$: plant state at $t = 1$

Now use the dynamics.

$$x(1) = Ax(0) + Bu(0).$$

1. Compute $Ax(0)$.
2. Compute $Bu(0)$.
3. Add the two terms to obtain $x(1)$.

Answer:

$$x(1) = \begin{bmatrix} \_\_\_\_\_\_ \\ \_\_\_\_\_\_ \end{bmatrix}$$

Task 3: One more NN step

Now repeat the NN computation using $x(1)$.

1. Pass $x(1)$ through the first layer:
$$z_1^{(1)} = W_1 x(1) + b_1.$$
2. Apply ReLU:
$$y_1^{(1)} = \text{ReLU}(z_1^{(1)}).$$
3. Compute the output:
$$u(1) = \text{ReLU}(W_2 y_1^{(1)} + b_2).$$

Answer: $u(1) =$ ____________

Task 4: Plant update to $x(2)$

Apply the dynamics once more:

$$x(2) = Ax(1) + Bu(1).$$

Answer:

$$x(2) = \begin{bmatrix} \_\_\_\_\_ \\ \_\_\_\_\_ \end{bmatrix}$$

Task 5: Write the execution

Write the execution of length two starting from $x(0)$:

$$x(0),\ x(1),\ x(2).$$

$$\eta = \begin{bmatrix} \_\_\_\_\_ \\ \_\_\_\_\_ \end{bmatrix}, \quad \begin{bmatrix} \_\_\_\_\_ \\ \_\_\_\_\_ \end{bmatrix}, \quad \begin{bmatrix} \_\_\_\_\_ \\ \_\_\_\_\_ \end{bmatrix}.$$